An improved feature extraction method for self-paced brain-computer interface application

CHEN Guangming, ZHANG Jiacai, YAO Li

Abstract—**EEG based motor imagery is widely used in most practical Self–paced brain-computer interface systems where the user conveys his/her intents at will whenever they wish to do so, and the system doesn't tell the user when to perform the mental tasks that convey their intents to the system. The most important step for the Self-paced BCI system is to discriminate between characteristic mental activity changes and ongoing EEG, this step is used to detect the motor imagery task state from on ongoing brain activity, and determine the type of motor imagery task when motor imagery related brain activity is detect. In practical application of online signal processing for SBCI systems, numerous features can be extracted from EEG signals, but only a subset of them are really discriminative. In this case, a better feature extraction and selection method becomes critical and leads to many good performance for SBCIs. This paper shows an improved feature extraction and selection method, features are extracted by stationary wavelet transform and bandpass filtering, and support vector machine classify these extracted feature vectors. Our method selects the EEG features and classifier parameters by genetic algorithm. We test our methods in the BCI competition 2008 dataset I, our informal results indicate that our method is efficient for feature extraction and selection in Self-paced BCI system.**

I. INTRODUCTION

Motor imagery SBCI (self–paced brain-computer interface) system allows users to be more convenient to control the outside devices or computers using a specific type of mental activity, instead of normal muscles normal output channels[1]. It is widely accepted in the brain computer interface (BCI) research community that self-paced BCI is far more natural to use [2]. Unlike Most existing brain-computer interfaces (BCIs) which detect specific mental activity in a so-called synchronous paradigm and are operational at specific system-defined periods[3], self-paced (asynchronous) interfaces should only activates when the user intends to control, and should remain inactive at all other times. The state when a user is intentionally attempting to output a control command is called an intentional control state (IC), and the other times SBCI system is in a no-control (NC) state [2], [4].

SBCI is more useful, robust, operable, and deserve more attention, however, implementation of SBCI is much more difficult than a traditional synchronized BCI system [5]. In this paper, we focus on EEG signal processing in SBCI, and train the BCI system to detect those brain patterns related to the intentional control in the ongoing EEG. Signal processing in SBCI is characterized in the additional step to discriminate between the IC and NC mental brain activity. The performance of SBCI depends on this discrimination of IC and NC heavily.

The main concern in this paper is how to distinguish the mental brain activity related to motor imaginary (MI) effectively. A lot of feature extraction methods are applied in the research of brain-computer interface (BCI), e.g. bandpass filter, wavelet transform, and common spatial patterns (CSP) [4]-[7]. We notice that most of the methods extract time-frequency features and fewer papers make comparison of these methods. But the parameters for time/frequency filters design are always determined by prior experience [5]. To find better features, genetic algorithm (GA) is used to select these features. GA is not the first time be used in SBCI system [2]. But application of GA algorithm always only cared about how to use this method to choose the feature from the rhythms based on prior knowledge, such as the selection of beta/mu rhythm. Although GA is developed to select the features, but where filters group are determined and a specific filter is bind with a classifier [2]. We want to utilize information as much as possible, and we do not care about the prior knowledge. Any rhythm, channel, and time period is all included in our candidate feature database. All features selection is just decided by GA. We want to know whether the result be better, if some feature always think as unrelated with MI by prior experience can work with the related ones together.

In this paper, we try to built a candidate feature database of time-frequency features and choose the better features for the partition between the IC and NC mental brain activity. Many features are extracted from different periods and frequency for each EEG channel by filter groups. The candidate filter groups include match filter and stationary wavelet transform (SWT). These outputs of these filters are built as a feature database at first, then a GA based methods is proposed to identify the useful ones. And thus SBCI system is improved by designing classifiers from this subset of features.

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The reminder the paper is organized as follows: section II shortly describes the details of the application of GA methods in EEG features selection and parameters optimization. Then, the results are illustrated in section III, and we conclude with its discussion in section IV.

II. METHODS AND MATERIAL

A. Data Recording

The imagery data sets I of BCI competition IV are provided by the Berlin BCI group [8], downloaded from http://ida.first.fraunhofer.de/projects/bci/competition_iv/. The recording was made using BrainAmp MR plus amplifiers and a Ag/AgCl electrode cap from healthy subjects without feedback in the whole motor imagery session. For each subject two classes of motor imagery were selected from the three classes (left hand, right hand, and foot).

The data sets are from seven subjects. For each subject, the Berlin group provides two kinds of data sets. One is calibration data, the other is evaluation data. For the calibration data, arrows pointing left, right, or down were presented as visual cues on a computer screen. Cues were displayed for a period of 4s during which the subject was instructed to perform the cued motor imagery task. These periods were interleaved with 2s of blank screen and 2s with a fixation cross shown in the center of the screen. These data sets are provided with complete marker information. For the evaluation data, the motor imagery tasks were cued by soft acoustic stimuli (words "left", "right", and "foot") for periods of varying length between 1.5 and 8 seconds. The end of the motor imagery period was indicated by the word "stop". Intermitting periods had also a varying duration of 1.5 to 8s. The maker information is supplied after the end of the BCI competition IV.

All the signals are continuous signals of 59 EEG channels, which were measured most densely distributed over sensor motor areas and band-pass, filtered between 0.05 and 200 Hz.

B. MI Related Brain Activity

The time-frequency information is useful for the SBCI system based on motor imaginary (MI) activities in EEG [7]. When motor imagery happen, changes in the power of some rhythms, e.g. Beta/Mu rhythms, will be find and these changes are locked to some special time and localized in some electrodes. These MI related potentials changes provide information to explore the cognitive functions of the brain [9]-[11]. Here we consider the time, frequencies and spatial patterns in EEG.

However, these useful rhythms, for example, Beta/Mu rhythms have been proven related to motor imagery, they still have a wide range. We need to find more discriminate features from these rhythms. Many papers give different frequency bands, as follows: 14-18Hz, 18-22Hz, 22-26Hz, 22-30Hz, 14-30Hz, 26-30Hz. On the other hand, these changes take place in some special time period. MI activity changed in different time window of EEG trial, an appropriate time period which may contains more information. And thus SBCI system should pay more attention to those EEG segment with more obvious brain patterns. In the practical application, the choice of EEG channels is worth considering. A lot of papers provide some evidence that data from channel C3 and C4 are useful for discriminate [12], [13].

C. MI Related Candidate Features

EEG from all channels will be filtered by the filter group and a time window is used for getting a special time period. We do these to explore information as much as possible. But data will be too large to be disposed. To reduce the dimensionality of the feature space, variance of data from a special rhythm and time period is gained as a feature instead of EEG in the rhythm and time period. The figure 1 shows how to built features database.

To get frequency information, FIR filter are usually to find appropriate rhythm. Different frequency band have different effect on the partition of mental brain activity. Mu rhythm and Beta rhythm are used widely. Here, we design a filter group to deal with EEG data from each channel. It contains 4 FIR filters for different rhythms (0-5Hz, 7-13Hz, 11-17Hz, 18-22Hz).

Another method to get frequency information is discrete wavelet transform (DWT) [14]. DWT is a powerful tool for extracting time-frequency features. But the scale and translation parameters in wavelet function affect the time-frequency analysis results. For DWT is shift variant, and the values of wavelet coefficients may vary even with small shifts in time [14], so stationary wavelet transform (SWT) are choose, for it is shift-invariant and easy to use. The filters group contained 3 different continuous wavelet (morl, db4, sym). For the continuous wavelet filters, we also select data from different levels after wavelet transform e.g. data from 2 to 8 levels after morl wavelet transform. The sums of data from different levels are normalized as the out of the filter.

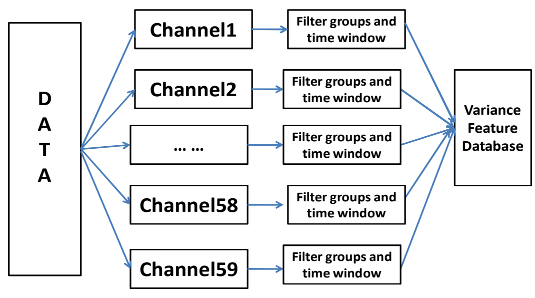

Fig. 1. MI Related fetures are candidate for SBCI systems.

D. MI Related Features Selection by GA

To find out an appropriate subset from the candidate MI related features database is an optimization problem. For this problem [2], many feature selection methods, such as exhaustive search, forward searching, and backward searching, is not practical due to huge computational loads or have the problem of the local minima. The genetic algorithm can better solve these problems. In the train step, we used the genetic algorithm-based feature selection method, and the feature subset found is used as the input of classifier.

Aiming at finding a subset of features out of the extracted MI features, we use genetic algorithm and take the classifier accuracy as the evaluation function. All the feathers of interest are coded in the form of a binary string which is called chromosome. The bits of the chromosome specify whether or not feature is selected by the GA. A value of "1" indicates the presence of the feature and "0" indicates the absence of the feature in the chromosome. The first chromosome is randomly generated. In my experiment, the chromosome holds 74 bits. The first 59 bits are used to decide which channels are choose, the following 7 bits are related with the choice of filters, and the last are the parameter needed by SVM. The number of the chromosome is 200 and the max generation is limited in 50.

E. *Two States (IC/NC) Detection*

In SBCI systems, user can output an intentional control at any time, which may output a control command to outside environment with different time period. Online SBCI should continue to detect the brain patterns and EEG segmentation is inevitable. But how to choose the window length for a decision is another key for SBCI system.

So an appropriate time window would affect the last classification result. We firstly use the time window with length of 4 seconds, for the calibration data can easily supply data for training in a period of 4 seconds. However the window may be too long to detect the change in some situation. An improved method is applied that EEG data from each channel is slipped by a time window with the length of 4 seconds, but the step is 0.5 second. We also use the time window length of 1 second and the step is 1 second. This time window will divide the 4 second calibration data into 4 parts. The figure 2 shows EEG is slipped by a time window with the length of 4 seconds and the figure 3 shows how to be slipped by a time window with length of 1 second.

F. *EEG Classification*

EEG signals of filters from slipping time window didn't input to the classifier directly, their variances are calculated and combined to construct the input feature vector of classifiers.

In order to classify the periods in which the user has no control intention and intentional control. From the EEG signals in time window, variances from all selected filters make up a features set for each channel.

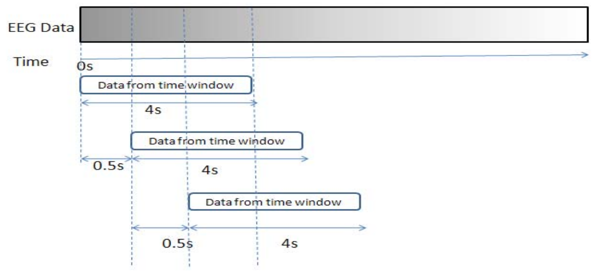

Fig. 2. EEG Data is slipped by 4 second time window.

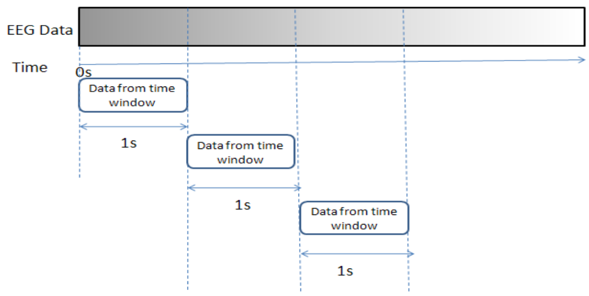

Fig. 3. EEG Data is slipped by 1 second time window.

A lot of classification methods are used in BCI and SBCI system. For example: linear Mahalanobis distance classifier (LMD), quadratic Mahalanobis distance classifier (QMD), Bayesian classifier (BSC), multi-layer perception neural network (MLP), probabilistic neural network (PNN), and support vector machine (SVM)[3], [15]. The SVM is a relatively new classification technique developed by Vapnik which has shown to perform strongly in BCI. SVM can avoid over-fitting. SVM can get a better solution than the other approaches and is widely used by many papers.

Here, we trained two SVM classifiers. The first classifier outputs a binary value for each period of time, the output label "-1" means the period of time belong to intentional control(IC) and "1" means they belong to no control (NC). And the additional classifier determines the type of motor imagery task when MI related brain activity is detected by the first classifier. However, in this paper we just give the result from the first classifier, for the main focus on the partition between the IC and NC mental brain activity. The parameters needed by SVM are still choosing through GA. To implement SVM, LIBSVM software is used [16].

III. Results and Discussion

For competition purpose, the maker information of evaluation is supplied after the end of the BCI competition IV. Here we provide two results. One doesn't utilize the true labels published by the organizer of BCI competition, thus the train dataset is divided new train dataset and test dataset. The other utilizes the true labels, and evaluation dataset used as test dataset.

A. *Results without True Labels (Experiment I)*

The data in the first experiment is from subject A and subject B. The test data and train data are both from Calibration data. 20 trails are used as train data and 180 trails are used as test data. The window is 4 second length. To evaluate the performance, here gives the results of IC state detection accuracy from two subjects in table I.

We compare the results of feature selection (channels, and so on) by GA and only the two channels (C3, C4). The experiment highlights that our method can learn the low-dimensional matrices that represent different kind of class features and can give an excellent classification results. The figure 4 shows the channels are selected by GA.

TABLE I
THE RESULT ON BCI COMPETITION IV DATASET I

Subjects	Channel	filter	Correct rate
A	AF3 Fz FC3 FC6 CFC1 CFC4 C3 Cz C4 C6 CCP3 CCP1 CCP2 CCP4 CP3 CP2 CP6 Pz P2 PO2 O1 O2	Sym2	91.2%
B	AF3 F5 F3 F6FC4 CF C3 CFC1 CFC7 C3 C1 Cz C4 T8 PO2 O1	11-17hz Morl Db4 Sym2	84.5%
A	*C3 C4*	7-13hz 18-22hz	75%
B	C3 C4	7-13hz	68%

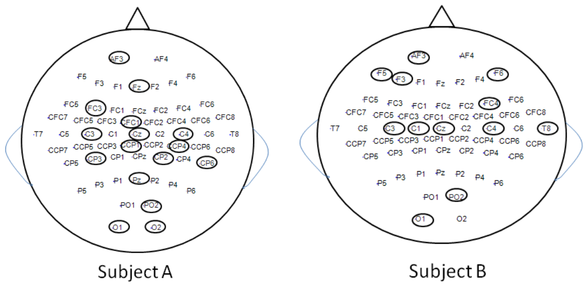

Fig. 4. The channels are selected by GA. Left figure is from subject A and right one is from subject B.

B. *Results with True Labels (Experiment II)*

The second experiment is done after the maker information is publicized. We use 100 trails from Calibration data as train data and the test data is 500 seconds long evaluation data. The slipping window length is 1 second length. The results of IC state detection accuracy from the subject A and B are in table II.

To see the different effect of time window, a contrast between 4 seconds time window and 1 second time window is give. We can see from the Table II, channel and filters and time window length should choose adjust to different person. The figure 5 shows the channels are selected by GA when data is slipped by different time windows.

TABLE II
THE RESULT OF DIFFERENT TIME WINDOW ON BCI COMPETITION IV DATASET I

Subjects	Time window	Channel	filter	Correct rate
A	1 second	AF3 AF4 F5 F1 FC1 FC6 CFC7 CFC3 CFC2 CFC6 T7 C2 C6 T8 CP2 PO1 O2	11-17 hz sym2	72%
B	1 second	AF4 F5 Fz F6 FC1 FC6 CFC5 CFC2 CFC6 C5 C4 C6 T8 CCP5 CCP3 CCP4 CCP6 CPz P6 PO1	morl db4	71%
A	4 second	AF3 F5 FCz FC2 CFC7 CFC3 CFC2 C4 C6 T8 CPz CP2 P3 O1 O2	7-13hz 11-17hz morl, db4	68%
B	4 second	AF3 F5 F3 CFC3 CFC1 CFC4 CFC6 CFC8 C5 C3 C2 C4 T8 CCP3 CCP4 CP3 CP2 P5 P3 P1 Po1 PO2 O2	morl db4 sym2	81%

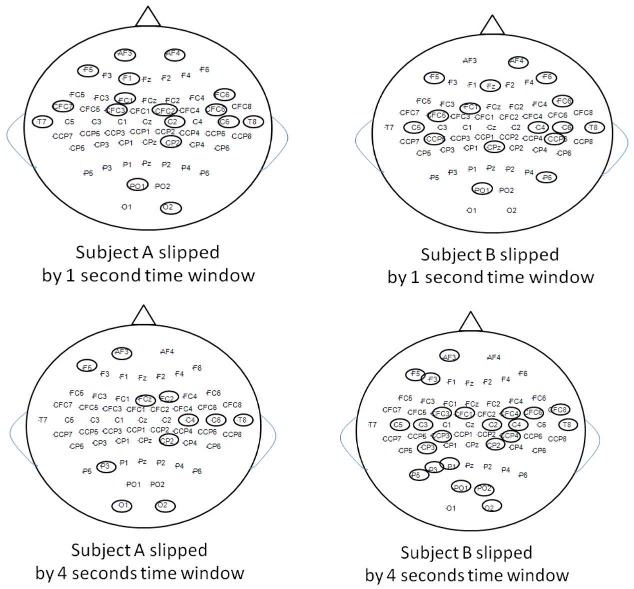

Fig. 5. The channels are selected by GA when data is slipped by different time window.

IV. CONCLUSION

This paper focuses on a new feature extraction method based on two filter groups to deal with the signal, these filters include stationary wavelet transform (SWT), and match filter, and then a small number of features are selected by GA.

In the feature extraction, SWT are better than match filter. As the table I show data from SWT is choose more than data from FIR filter though GA. The reason may be that SWT is more suitable for describing the changes of mind state and the features in FIR output is contained mainly in SWT output signals.

We can also notice that, our method has some adaption. Our signal processing algorithm should adapt to the user's signal features. For MI signal feature, the algorithm adjusts to the user's range of beta/mu-rhythm amplitudes and frequency, brain activity location and time window length.

However, MI features in EEG signals typically display time variations. Even for the same subject, the useful rhythm will vary with the time flow for physical condition varying. Of course, if the rhythm used by SBCI could be adaptive, the performance of SBCI system will be improved. Thus, effective SBCIs can't continue to be effective without the periodic online adjustments reducing the impact of such spontaneous variations. But in this case, different rhythm can work better together, and SWT work as a filter which weight different rhythms. In this paper, data from different levels after SWT work together to finish the weighting process. For GA selection a set of features, not a fixed frequency or channels are selected, and thus our method can work in more cases. Of course, the selection of better levels after SWT is important. Different wavelet use different levels. For example, we choose 3 level -6 level for the sym2 wavelet and 2 level -8 level for morl wavelet.

Our work also showed that the channels and rhythms unrelated with motor imagery are useful for classification of IC and NC. Some channels are thought related with motor imagery such as C3 and C4. These channels are got by GA which provides these channels are related with MI again. But we also find some channels unrelated with motor imagery are used to improve the result as the table I shows. We think that Electrode positions have a little shift for each experiment may cause different channel choose. These unrelated channels may provide assistant information as reference which helps to get a better result. To some extent, GA may also leads to this choose for it is suboptimal.

The first chromosome for GA can also affect our result heavily. Different initial value often causes results varying greatly. Considering computational loads, the number of the chromosome and the max generation for GA is less, so the not enough computation may cause this problem. Maybe the first chromosome is decided by some prior knowledge will save the computational time and has a better result.

Here we do not use the CSP, though CSP is widely used. Because in my other experiments, we find the CSP improve our technique less. The cause may be our technique and the CSP have the same effect. CSP can abstract important information by having the different channels weighted.GA do similar things. Based on GA searching, different channels are choose as channels are weighted by '0 'or' 1 '.

Time window is another thing we care about. As the second experiment show we can find the 1 second window have the same performance as the 4 second window to some extent. We use the 4 second window for the calibration data just supply the 4 second motor imagery time. If we use 1 second window, the full motor imagery time will be segmented in 4 parts. The contrast show appropriate partition of draft data do not reduce the performance. But through the contrast between experiment I and experiment II, we find the result in the experiment II is not better than it in experiment I. This may means that the best time window length should be further researched in the future.

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